UNIT 5 · STANDARD 6.GR.2

Determine the Volume of Rectangular Prisms

Lesson 5-5

Name: _____ **Date:** _____ **Class:** _____

CLASS LEGACY MISSION

Seal the Time Capsule

You are the lead builder for the school's 50-year time capsule project. Every memory must fit inside rectangular containers, and the principal needs to know exactly how much each one holds before they are buried. Master volume and the capsule gets sealed for the future.

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the volume of a rectangular prism with whole-number edges using length × width × height.

Language Objective: I can explain my work using the words volume, rectangular prism, cubic units, and edge.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Volume · **Rectangular prism** · **Cubic units** · **Length, width, height** · **Net** · **Edge** · **Unit cube** · **Base area**

Level 3: try the blanks from memory first, then check the bank.

1. The amount of space inside a three-dimensional solid is its _____.

2. A solid with six rectangular faces, like a box, is a _____.

3. The unit used to measure volume, like cubic centimeters, is _____.

4. The three edge measurements of a rectangular prism are its _____.

5. A flat pattern that folds up into a solid is a _____.

6. A line segment where two faces of a solid meet is an _____.

7. A cube that is one unit on every edge is a _____.

8. The area of the bottom face of a prism is the _____.

Write About the Math

How many more cubic inches does the large box hold than the medium box?
¿Cuántas pulgadas cúbicas más tiene la caja grande que la caja mediana?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Volume** *Volumen*
- ☐ **Rectangular prism** *Prisma rectangular*
- ☐ **Cubic units** *Unidades cúbicas*

- [ ] **Length, width, height**
  Largo, ancho, altura
- [ ] **Net**
  Plantilla (desarrollo plano)

Model: *Volume measures space in three directions, so you need length, width, and height.* — Students explain that volume is three-dimensional, so all three edges (length, width, height) are needed; two measurements only give area of one face.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The large box volume is ___ in³ because V = ___ × ___ × ___. The medium box volume is ___ in³. The large box holds ___ more cubic inches.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.